QBitSync

Math Club DC MiniProject 26'

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MC DC Open House 2026

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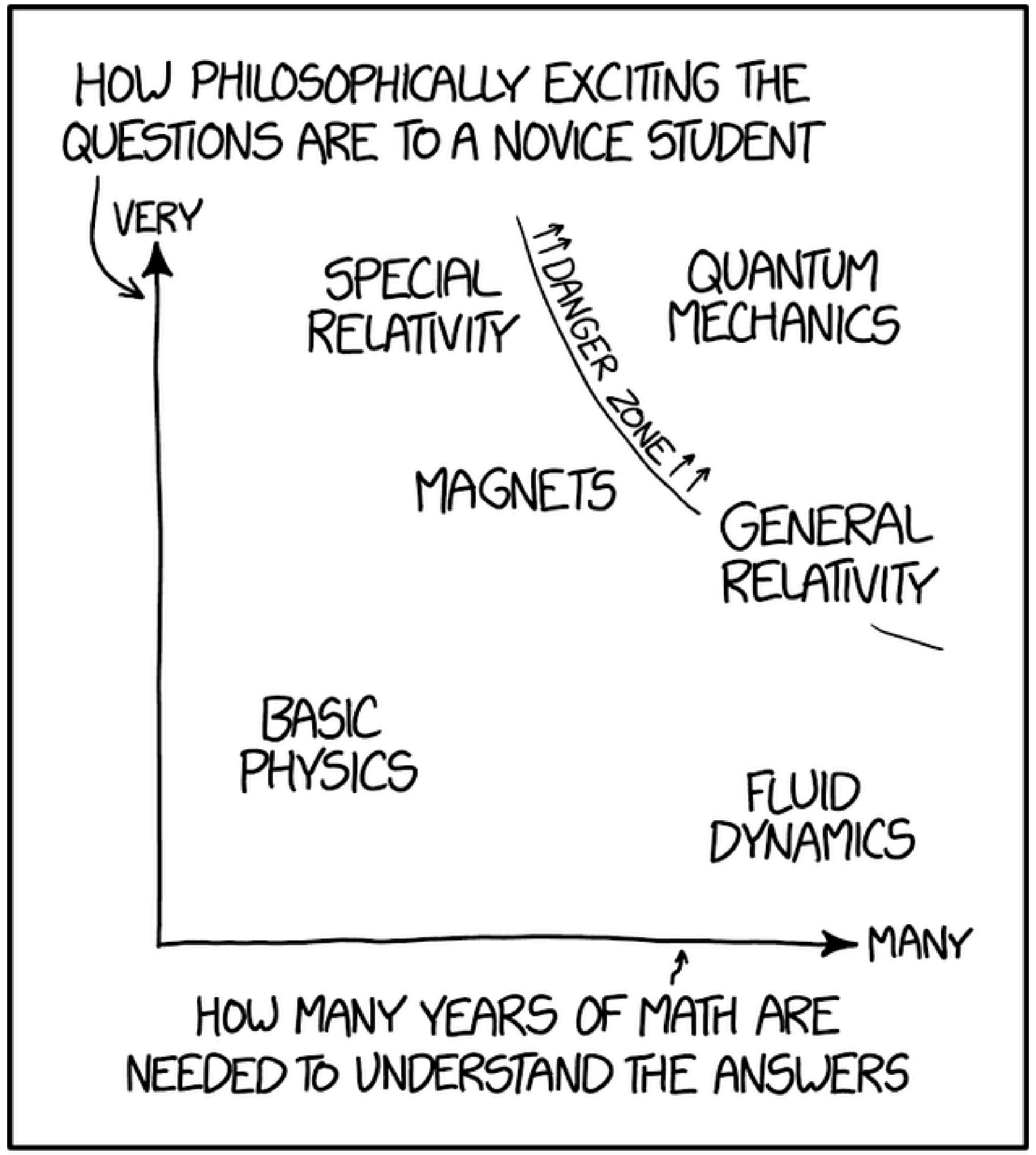
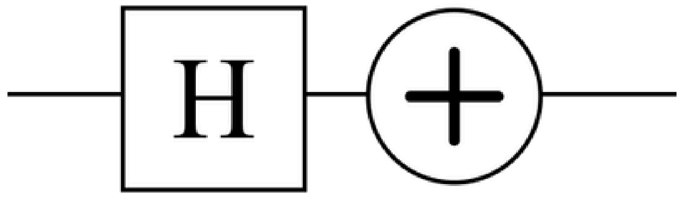
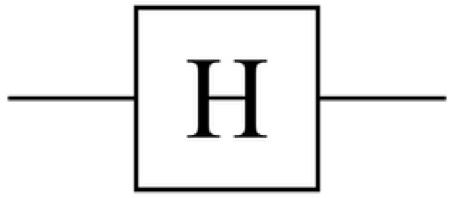
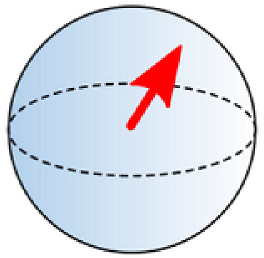
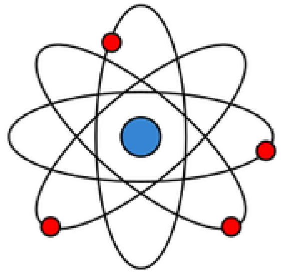


WHY SO MANY PEOPLE HAVE WEIRD IDEAS ABOUT QUANTUM MECHANICS

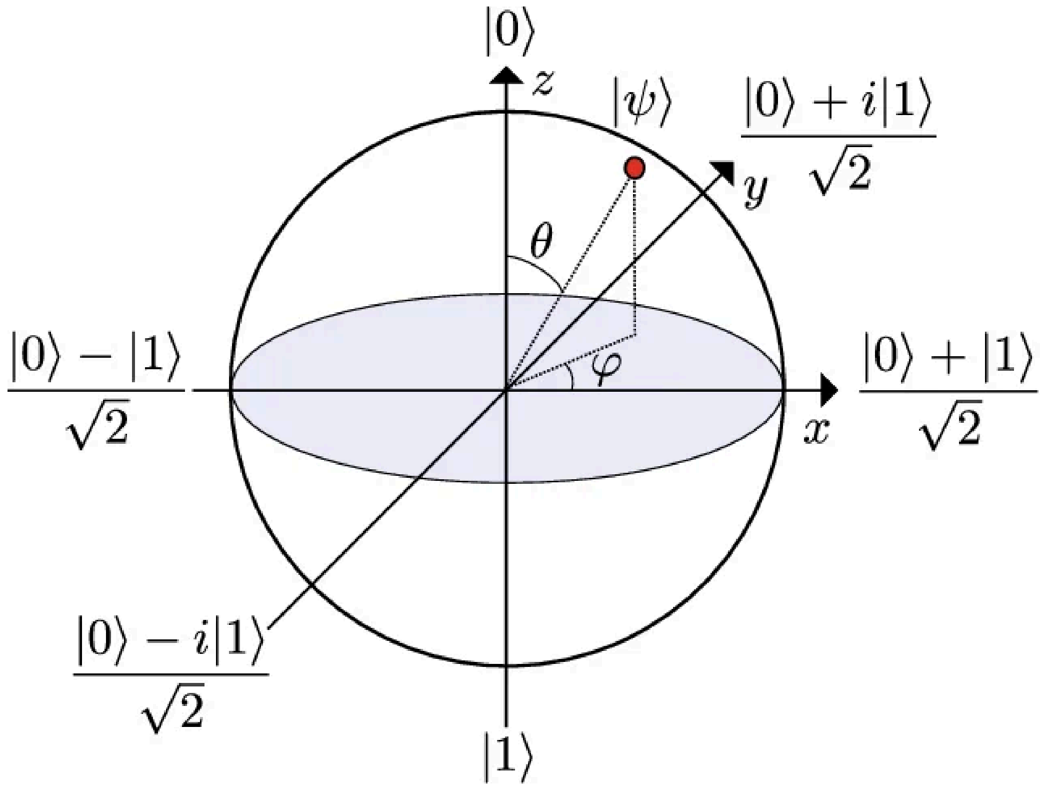
Intro

5	program	instruction sequence	`h q;` `x q;`
4	algorithm	quantum circuit	
3	unit of processing	quantum gate	
2	unit of information	qubit	
1	physical principle	quantum mechanics	

BLOCH SPHERE AND DIRAC NOTATION

Bloch Spere

It is a way of visualising quantam states. It is done using a unit sphere. The North Pole represents $|0\rangle$ and The South Pole $|1\rangle$. As you move from North to South, the probability of the state collapsing to 1 increases.



Dirac Notation

Dirac Notation (also known as Bra-ket Notation), is handy notation to represent vectors and their transpose.

Ket: $|\psi\rangle$ is a state vector

Bra: $\langle\psi|$ represents the conjugate transpose

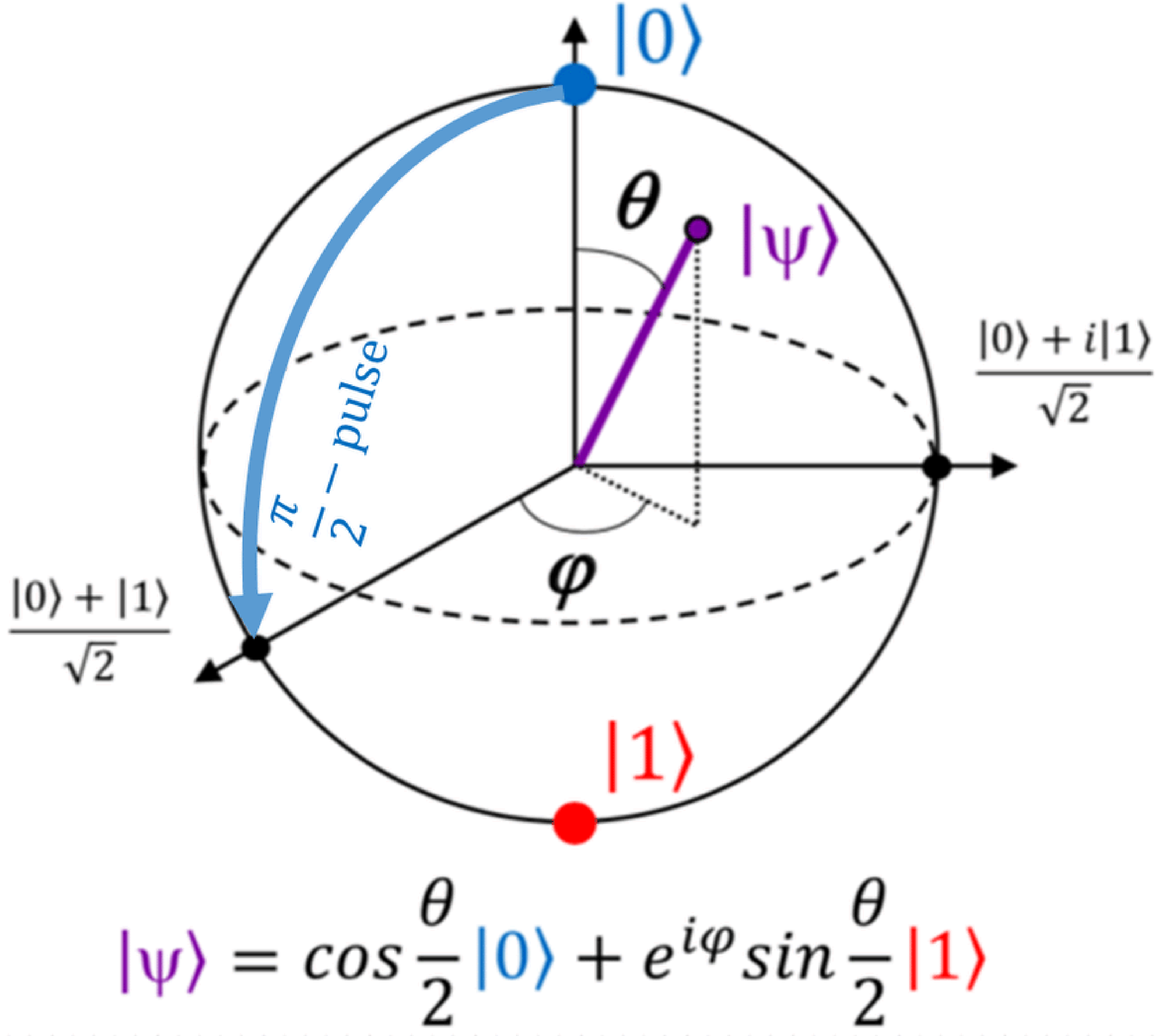
The above image is a visual of the bloch sphere, which will be explained in the next slide

Bloch Sphere

Every qubit only needs two numbers to represent it: relative phase as well as colatitude.
Any quantum state $|\psi\rangle$ can be written as

$$|\psi\rangle = cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}sin\left(\frac{\theta}{2}\right)|1\rangle$$

Where ɵ is the colatitude and ϕ is the relative phase. As you can see, the probability of measuring a certain state only depends on ɵ as the magnitude of $e^{i\phi}$ is 1. Every point on the Bloch Sphere represents a unique qubit (up to global phase as multiplying the above expression with some phase only rotates the universe and does nothing to the qubit itself.

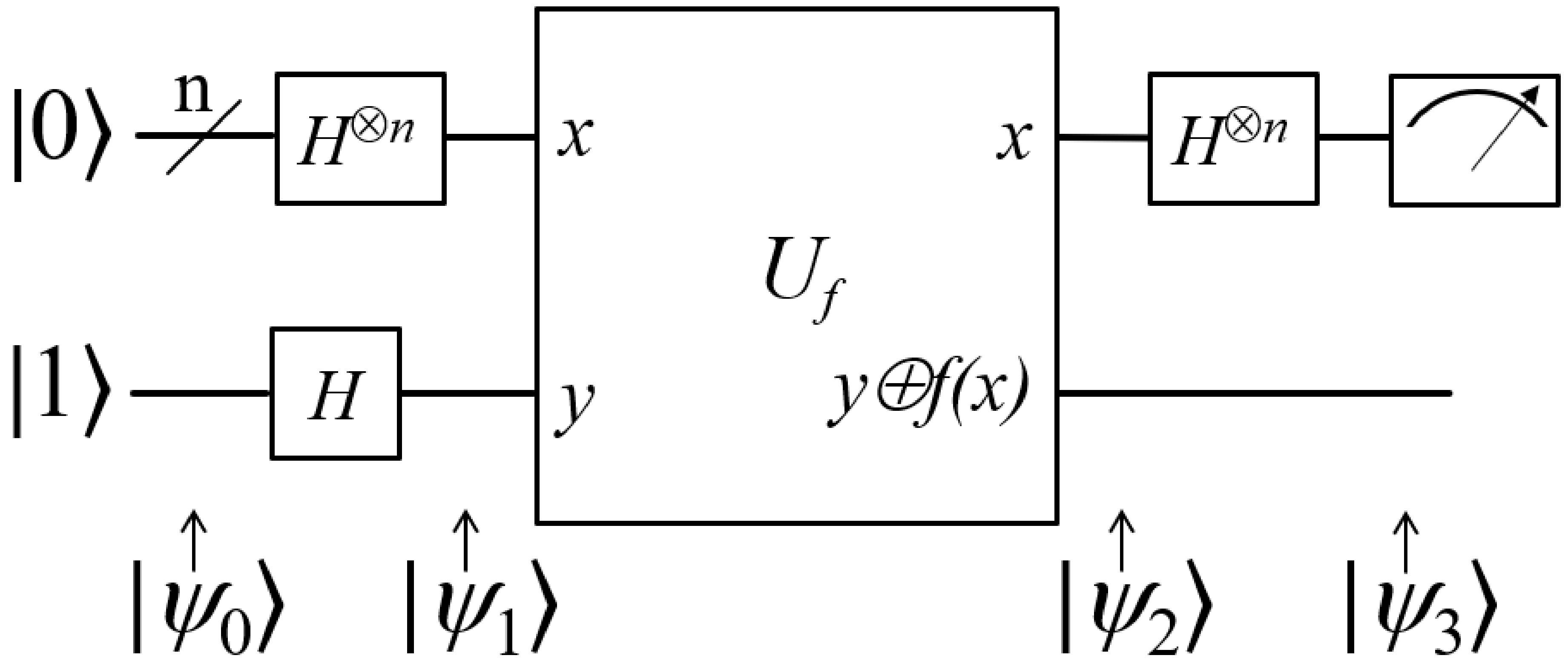


important points on the sphere:

- North Pole: $|0\rangle$
- South Pole: $|1\rangle$
- Equator: Probability of measuring 1 and 0 is the same. Two points on the equator are $|+\rangle$ and $|-\rangle$ which represent
  $\frac{(|0\rangle+|1\rangle)}{\sqrt{2}}$ and $\frac{(|0\rangle-|1\rangle)}{\sqrt{2}}$ respectively.

Quantum Algorithms

Quantum algorithms are computational methods designed for quantum computers, using superposition and entanglement to solve certain problems faster than classical algorithms. They are central to quantum computing, with applications in cryptography, optimization, and simulation.



Deutsch-Jozsa Algorithm

An algorithm that speeds up an $O(2^n)$ computation of finding whether a function is constant or balanced into a constant O(1) computation.

Bernstein-Vazirani Algorithm

Instead of an O(n) algorithm to find a binary string s, and the function f being a dot product of x and s, this again uses a O(1) computation.

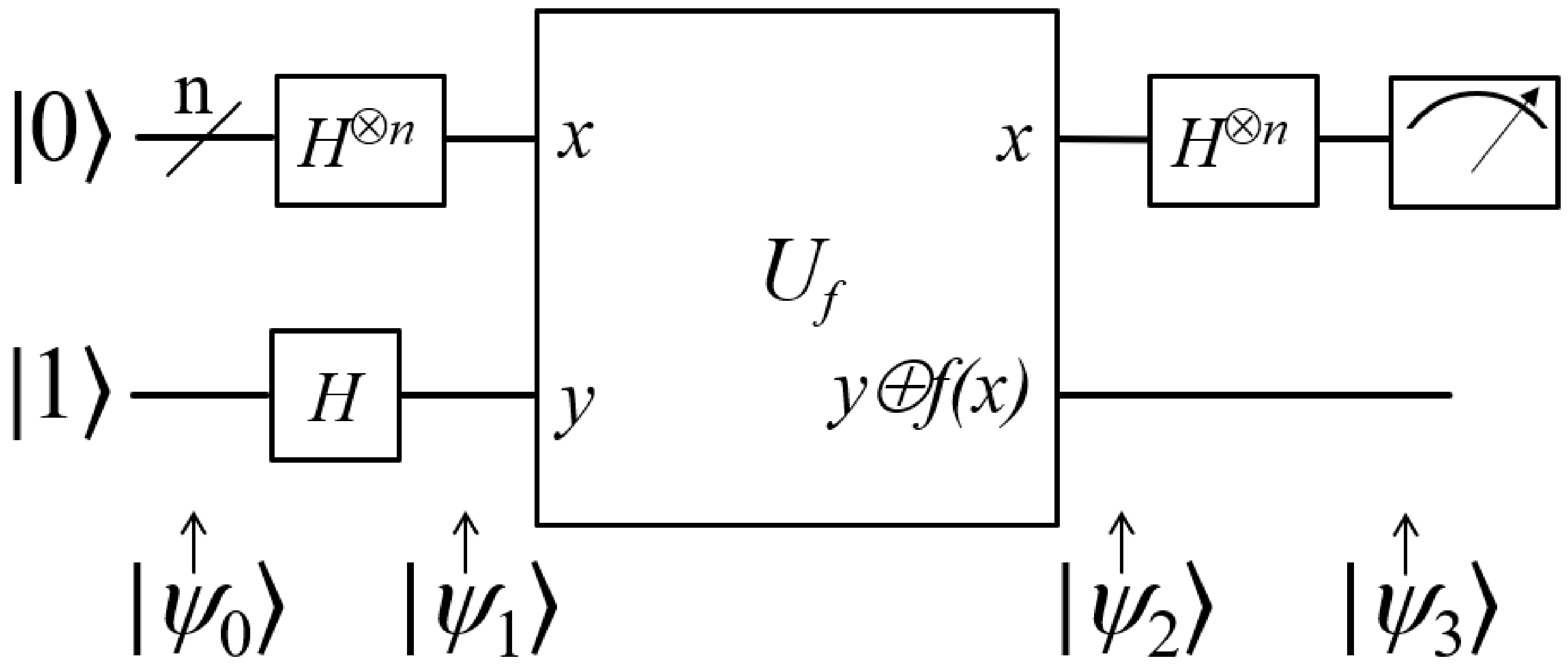
Shor's Algorithm

A fast and reliable quantum algorithm to factor large primes. This algorithm is very relevant to modern cryptography. (Explained later)

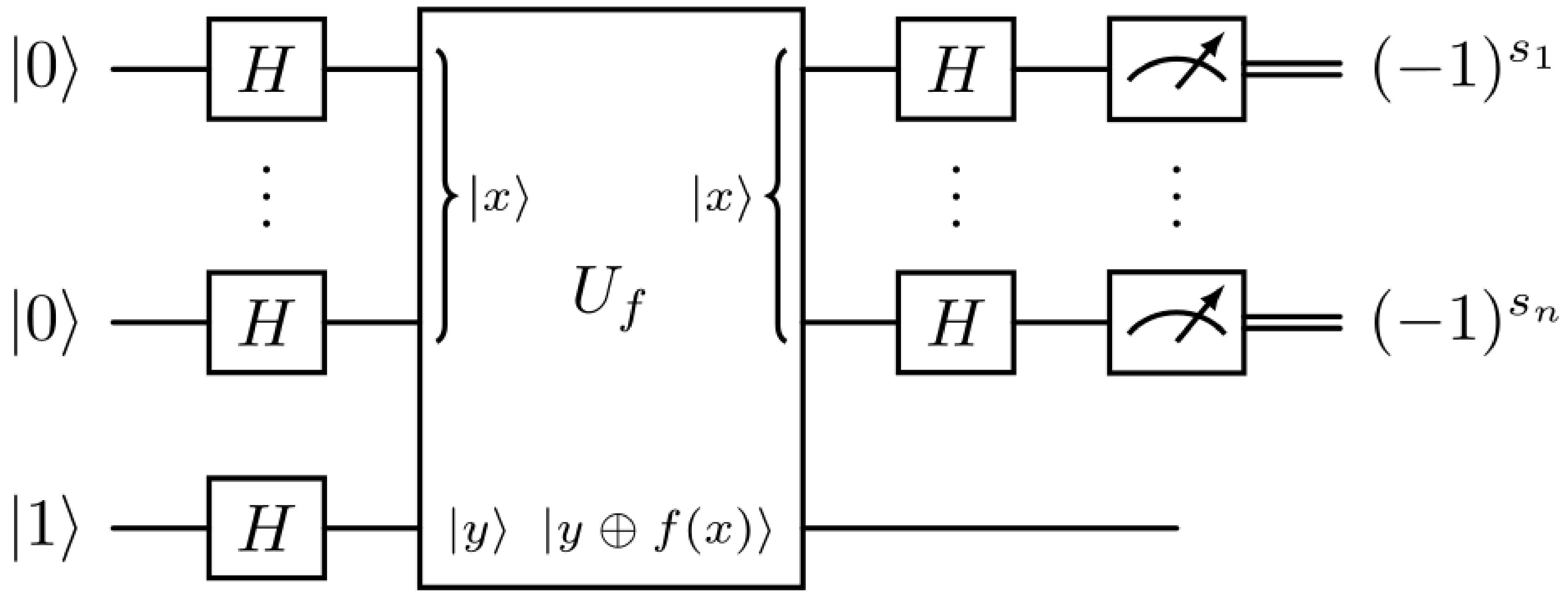
Deustch-Josza Algorithm

- This algo solves the problem of finding whether a given function f is **balanced** or **constant**.
- In a classical algorithm, it takes a maximum of $\mathbf{2^{n-1} + 1}$ queries to solve the problem.
- Quantum algorithms can solve this in a single run of the function f.
- A simplified rundown is given below:

1. The state starts as $|0\rangle^{\otimes n}|1\rangle$.
2. For the $|0\rangle^{\otimes n}$, a Hadamard gate transforms the bit into a $|+\rangle^{\otimes n}$ state, and the $|1\rangle$ is transformed into a $|-\rangle$.
3. The quantum oracle for the function f is U_f, and it is applied to the input, giving a singular output.
4. Applying a Hadamard to the first bit, and measuring it gives the result:
   a. $|1\rangle$ → Balanced
   b. $|0\rangle$ → Constant



Bernstein-Vazirani

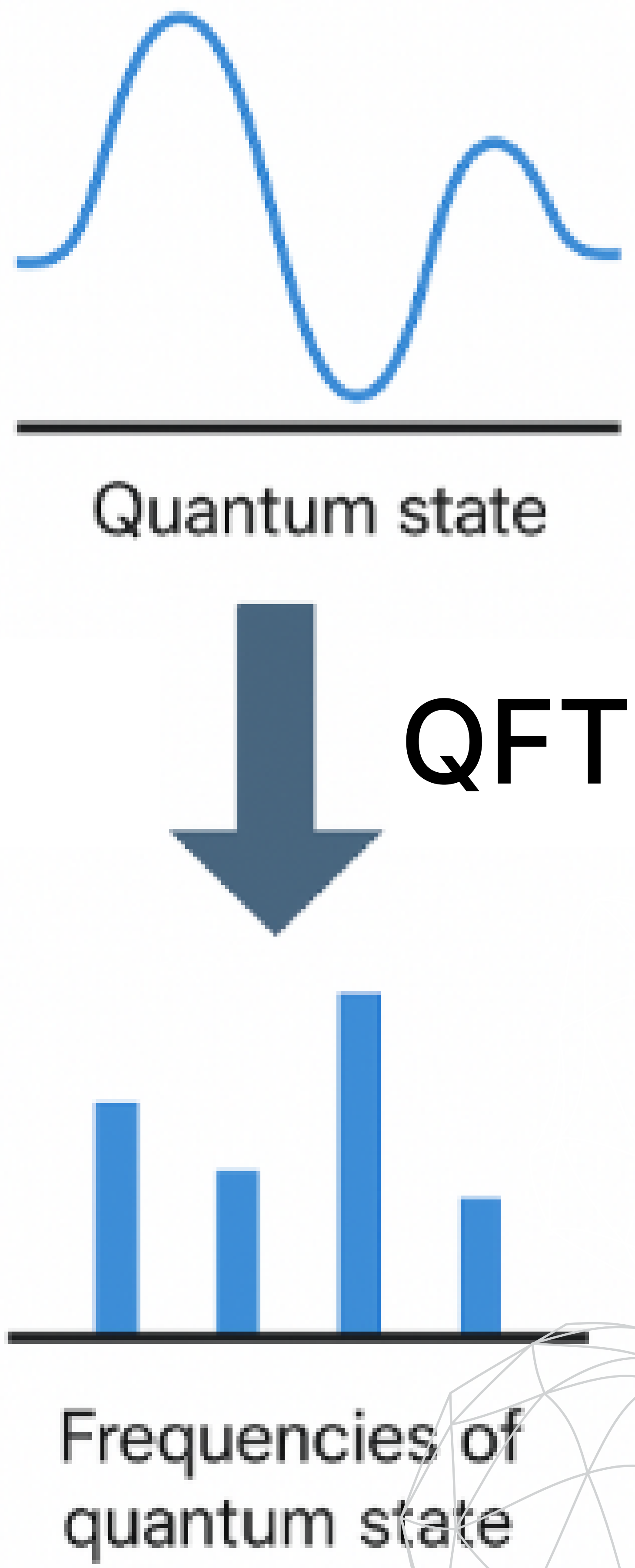


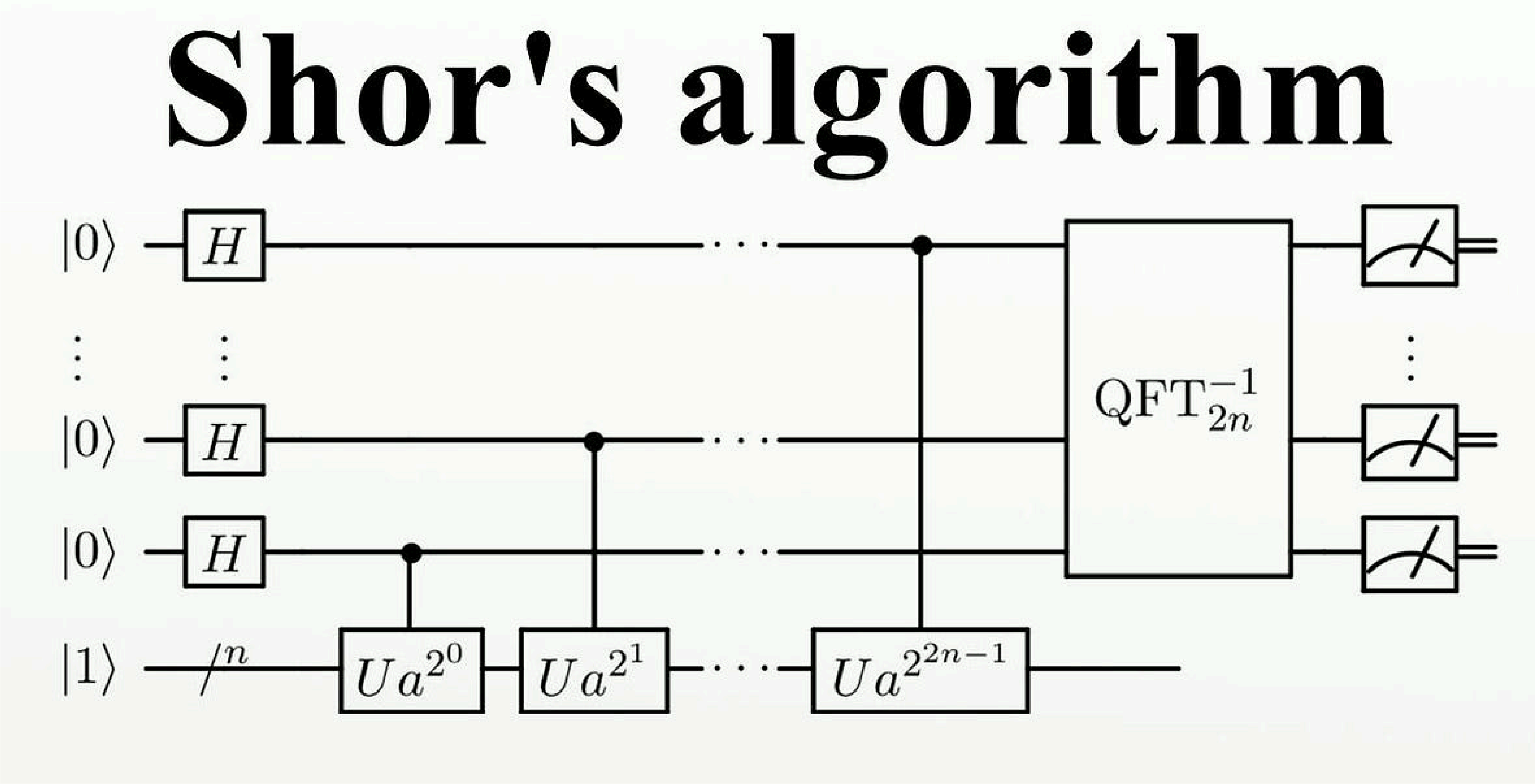
- This algorithm finds a secret binary string s such that a mystery function f(x) returns the dot product of x and s.
- In classical computing, it takes n runs of inputting 2^i every time to figure each string.
- Here, it takes a simple one run.
- The working is explained below:

1. The state starts as $|0\rangle^{\otimes n}|1\rangle$
2. For the states $|0\rangle^{\otimes n}$, a Hadamard gate transforms it into $|+\rangle^{\otimes n}$ states, and the $|1\rangle$ is transformed into a $|-\rangle$.
3. The quantum oracle for the function f is U_f, and it is applied to the input, giving n outputs.
4. Applying a Hadamard to each bit, and measuring it gives the result of that particular bit in the string s.

Quantum Fourier Transform

- Quantum equivalent of DFT
- DFT applied to amplitudes of the full multi-qubit state
- Inputs - qubit state
- Outputs - another qubit state, whose amplitudes are Fourier coefficients of input state
- Works using interference
- Exponentially faster then DFT
- Used in many quantum algos like QPE, Shor’s etc





QPE Ciruit used in Shor's Algorithm [$U_a(x) = ax(modN)$]

- It is used to find out the prime factors of a number N.
- It does so in approximately $\mathcal{O}(log^3(N))$ time which is much faster than all its classical counterparts [$\mathcal{O}(e^{log^{1/3}(N)})$].
- It mainly relies on quantum phase estimation to find the period/order of a number wrt N.
- It then uses classical techniques (continued fractions, euclidean theorem) to obtain a factor of N.

Applications :

- Many widely used cryptographic algorithms like RSA rely on the extreme classical difficulty of factoring large numbers. Shor's algorithm can be used to bypass them.

Thank You For Joining Us

Quantum computing is a vast growing field we have only scratched the surface of what this subject entails.
We hope you never gonna give us up, never gonna let us down, never gonna run around, and desert us, never gonna make us cry, never gonna say goodbye, never gonna tell a lie and hurt us.